Rational periodic points of monic integral conservative Hénon maps

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Abstract

We classify the rational periodic points of $H_{a,b}(x,y) = (y, y^2 + by + a - x)$ for every pair of integers a, b. All such points are integral, and their exact periods belong to $\{1,2,3,4\}$. The classification gives explicit coordinate words and parameter conditions for every orbit, including all coincidences and intersections of the parameter families. Integer translations reduce the problem to two normal forms, according to the parity of b. For all sufficiently negative normalized parameters, a maximum-coordinate bound restricts the orbit to six symbols on either an integer or a half-integer lattice. An exact coefficient separation produces the same local symbol equations in both cases, which we classify without a bound on the period. Thirty remaining parameters are covered by complete finite partial-permutation certificates. It follows that each map has at most eight rational periodic points. This bound is attained exactly when $b = 2q + 1$ and $a - q^2 + q = -4$ for an integer q. We also give the ordinary-time rational-point return counts and their finite-orbit zeta functions. The coefficient-integrality, monicity, and Jacobian-one hypotheses are retained throughout.

1 Introduction and classification

The rational periodic set of a polynomial automorphism can be finite without admitting a usable description as the coefficients vary. We consider the complete classification problem for the family

$$H_{a,b}(x,y) = (y, y^2 + by + a - x), \qquad a, b \in \mathbb{Z}. \tag{1.1}$$

The determinant of its Jacobian is $+1$. Both the sign of the final term and the condition that the quadratic coefficient is exactly one are part of the problem. For a map H, write

$$\mathrm{Per}(H, \mathbb{Q}) = \{P \in \mathbb{Q}^2 : H^n(P) = P \text{ for some } n \geq 1\}.$$

The exact period of P is the least such positive integer n. All times in this article refer to ordinary iteration of H.

A finite coordinate word $(v_0, \ldots, v_{d-1})$ denotes the orbit whose successive points are (v_i, v_{i+1}), with subscripts read modulo d. Words are identified under cyclic rotation, but not under an arbitrary permutation of their entries. The word describes an orbit of $H_{A,e}$ precisely when

$$v_{i-1} + v_{i+1} = v_i^2 + ev_i + A \quad \text{for every } i. \tag{1.2}$$

Repeated coordinates are allowed; the least period of the resulting orbit must still be checked.

Theorem 1.1 (Complete rational periodic set)**.** *Let $a, b \in \mathbb{Z}$. Write uniquely $b = 2q + e$, where $q \in \mathbb{Z}$ and $e \in \{0, 1\}$, and put*

$$A = a - q^2 + (2 - e)q. \tag{1.3}$$

Every rational periodic point of $H_{a,b}$ is integral. The complete list of its periodic orbits is obtained from Table 1 when $e = 0$, or Table 2 when $e = 1$, by subtracting q from every coordinate in the listed normalized words. At a parameter appearing in several rows, take the union of those rows, counting coincident orbits only once.

Every exact period belongs to $\{1, 2, 3, 4\}$, and

$$\#\,\mathrm{Per}(H_{a,b}, \mathbb{Q}) \leq 8. \tag{1.4}$$

Equality holds if and only if $e = 1$ and $A = -4$, equivalently $b = 2q + 1$ and $a - q^2 + q = -4$.

Parameter A	Coordinate words	Exact period and range
$1 - k^2$	$(1 - k)$, $(1 + k)$	1; $k \geq 0$; coincide at $k = 0$
$-k^2 - 3$	$(-k - 1, k - 1)$	2; $k \geq 1$
$-k^2 - 1$	$(-k - 1, k, k)$, $(k - 1, -k, -k)$	3; $k \geq 0$; coincide at $k = 0$
$-k^2$	$(-k, -k, k, k)$	4; $k \geq 1$

Table 1: All rational periodic orbits of $H_{A,0}$, written as cyclic coordinate words. The two coincidence clauses apply only to the two words in their own row. At overlapping parameters the table means their union; all ranges are part of the classification.

Parameter A	Coordinate words	Exact period and range
$-k(k + 1)$	$(-k)$, $(k + 1)$	1; $k \geq 0$; always distinct
$-k(k + 1) - 4$	$(-k - 2, k - 1)$	2; $k \geq 0$
$-k(k + 1) - 2$	$(-k - 2, k, k)$	3; $k \geq 0$
$-k(k + 1) - 2$	$(k - 1, -k - 1, -k - 1)$	3; $k \geq 1$
$-k(k + 1) - 1$	$(-k - 1, -k - 1, k, k)$	4; $k \geq 0$

Table 2: All rational periodic orbits of $H_{A,1}$. The second three-cycle row starts at $k = 1$: its word at $k = 0$ would be the fixed point $(-1, -1)$, already present in the fixed-point row at parameter $A = -2$. Overlapping rows again mean their union.

The coefficient normalization, the shared symbol argument, and the exceptional parameters are all needed for this statement. A census at finitely many parameters would not exclude cycles at new parameters or of larger period. Conversely, a uniform period bound would not determine which points occur at each parameter. Here the large-parameter proof leaves an explicitly bounded finite complement, rather than treating a sample as evidence for the general case.

Relation to arithmetic finiteness and period bounds. The height-theoretic study of Hénon maps is classical; foundational work includes Silverman [5], and Ingram develops canonical-height and specialization results in [2]. Our proof uses a direct local maximum argument to establish integrality and does not claim a new general finiteness theorem. For arbitrary polynomial maps of $\mathbb{Z}^2$ with integer coefficients, Pezda [4, Theorem 2.1] determines the possible cycle lengths, with largest value 24. The task here is different: it asks for the exact point sets in a fixed quadratic automorphism family at every parameter.

Degree, coefficients, and determinant sign. Ingram's detailed conjecture for quadratic maps concerns $(x, y) \mapsto (y, x + y^2 + c)$ with $c \in \mathbb{Q}$, whose determinant is -1; see [2]. The same sign distinction is explicit in the number-field questions of [1, Section 11]. Theorem 1.1 does not resolve that conjecture, nor a uniform-boundedness conjecture for all rational-coefficient Hénon maps. In contrast, the constructions of Kim, Krieger, Postolache, and Szeto [3] do have determinant $+1$. Their long cycles arise in growing odd degree from rational-coefficient integer-valued polynomials, rather than from the monic quadratic integral-coefficient family (1.1).

These comparisons concern the precise objects and statements just described, not a claim of global priority. In particular, our access to Silverman [5] was limited to its publisher record; its arithmetic context is also discussed in the accessible introductions of [2, 3]. We make no assertion about examples or classification statements in its unread subscription text. The proof below is self-contained.

The argument proceeds through integrality and integer translations in Section 2, followed by a common six-symbol reduction in Section 3. Section 4 classifies the resulting local equations. Section 5 and Appendix A close the finite complement. Section 6 proves the sharp bound, and Section 7 records the ordinary-time return law and its scope.

2 Integrality and normal forms

The inverse of (1.1) is

$$H_{a,b}^{-1}(x,y) = (x^2 + bx + a - y, x).$$

Thus every periodic orbit gives a cyclic rational coordinate sequence $(x_i)_{i\in\mathbb{Z}/n\mathbb{Z}}$ satisfying

$$x_{i-1} + x_{i+1} = x_i^2 + bx_i + a. \tag{2.1}$$

All arguments below allow repeated coordinates and an arbitrary positive period n.

Lemma 2.1 (Integrality)**.** *If $a, b \in \mathbb{Z}$, every rational periodic point of $H_{a,b}$ belongs to $\mathbb{Z}^2$.*

Proof. Fix a prime p and let $M = \max_i |x_i|_p$. If $M > 1$, choose j with $|x_j|_p = M$. The term x_j^2 has norm M^2, whereas $|bx_j|_p \le M$ and $|a|_p \le 1$. The ultrametric inequality, with its unique dominant term, gives

$$|x_j^2 + bx_j + a|_p = M^2.$$

The left-hand side of (2.1) has norm at most M, a contradiction. Hence $|x_i|_p \le 1$ for every prime p and every i. A rational number integral at every prime is an integer. ☐

Lemma 2.2 (Integer translation)**.** *With q, e, A as in Theorem 1.1, the map $T_q(x,y) = (x+q, y+q)$ satisfies*

$$T_q \circ H_{a,b} \circ T_q^{-1} = H_{A,e}. \tag{2.2}$$

It preserves rationality, integrality, and exact periods. The original coordinates are obtained from normalized coordinates by subtracting q.

Proof. Put $x = u - q$ and $y = v - q$. The second coordinate after conjugation is

$$v^2 + (b - 2q)v + a + q^2 - bq + 2q - u = v^2 + ev + A - u,$$

and the first is v. The remaining statements follow because the translation and its inverse are defined over $\mathbb{Z}$. ☐

We henceforth use $x_i \in \mathbb{Z}$ for normalized coordinates and treat $e = 0, 1$. Summing (1.2) around the cycle gives

$$\sum_i (x_i - 1)^2 = n(1 - A), \qquad e = 0, \tag{2.3}$$

$$\sum_i (x_i - \tfrac{1}{2})^2 = n(\tfrac{1}{4} - A), \qquad e = 1. \tag{2.4}$$

For $e = 0$, there is no real periodic orbit when $A > 1$, and the case $A = 1$ consists only of the fixed point $(1, 1)$. For $e = 1$, every summand in (2.4) is at least $1/4$ because x_i is integral. Consequently no rational periodic orbit exists when $A > 0$. At $A = 0$, all coordinates are 0 or 1. A zero coordinate forces both neighbors to be zero, and a coordinate 1 forces both neighbors to be 1. There are exactly two fixed points, $(0, 0)$ and $(1, 1)$.

Both branches can be centered into the same recurrence. Set

$$\begin{array}{c|c|c|c} e & u_i & c & \text{coordinate lattice} \\ \hline 0 & x_i & -A & \mathbb{Z} \\ 1 & x_i + \frac{1}{2} & -A - \frac{3}{4} & \mathbb{Z} + \frac{1}{2} \end{array} \tag{2.5}$$

Then

$$u_{i-1} + u_{i+1} = u_i^2 - c. \tag{2.6}$$

The summation center $x_i - 1/2$ in (2.4) and the recurrence center $x_i + 1/2$ in (2.5) have different purposes; their signs are not interchangeable.

Lemma 2.3 (Maximum-coordinate bounds)**.** *For a real cyclic solution of* (2.6)*, let* $R = \max_i |u_i|$*. Whenever* $c \geq -1$*,*

$$R \leq 1 + \sqrt{1 + c}, \qquad |u_i^2 - c| \leq 2R \quad \text{for every } i. \tag{2.7}$$

Proof. At a maximum coordinate, the recurrence gives $R^2 - c \leq 2R$, hence $(R - 1)^2 \leq 1 + c$ and the first bound. At every index, the sum of the neighboring coordinates has absolute value at most $2R$, giving the second bound. □

We apply this lemma to the even branch with $A \leq 0$, and to the odd branch with $A \leq 0$. The case $A = 0$ in the latter branch will also appear in the finite certificate for uniform bookkeeping, although it has already been settled directly.

3 A common six-symbol reduction

For large negative normalized parameters, the residual bound in Lemma 2.3 leaves only three possible absolute coordinate values. The two lattices require different rounding arguments, which we keep explicit.

Lemma 3.1 (Integer centers)**.** *Suppose* $c \in \mathbb{Z}$*,* $c \geq 13$*, and a cyclic solution of* (2.6) *has all coordinates in* $\mathbb{Z}$*. There is a unique integer* $r \geq 4$ *such that*

$$r^2 - r + 1 \leq c \leq r^2 + r. \tag{3.1}$$

Put $s = c - r^2$*, so* $1 - r \leq s \leq r$*. Every coordinate has a unique representation*

$$u_i = \varepsilon_i r + \delta_i, \qquad \varepsilon_i \in \{-1, 1\}, \quad \delta_i \in \{-1, 0, 1\}. \tag{3.2}$$

Proof. The intervals in (3.1) begin at 13 when $r = 4$, and the next lower endpoint is $(r+1)^2 - (r+1) + 1 = r^2 + r + 1$. They therefore partition every integer $c \geq 13$.

If $s \leq -2$, then $c + 1 < r^2$. The first bound in (2.7), together with $R \in \mathbb{Z}$, gives $R \leq r$. If $s \geq -1$, then $c + 1 \leq r^2 + r + 1 < (r+1)^2$, and the same argument gives $R \leq r + 1$.

Suppose now that $|u_i| \leq r - 2$ at some index. In the first case,

$$u_i^2 - c \leq -4r + 4 - s \leq -3r + 3 < -2r \leq -2R,$$

where the strict inequality uses $r \geq 4$. In the second case,

$$u_i^2 - c \leq -4r + 5 < -2r - 2 \leq -2R.$$

Both contradict the pointwise bound in (2.7). Consequently $r - 1 \leq |u_i| \leq r + 1$. These are precisely the six distinct integers described by (3.2). $\square$

Lemma 3.2 (Half-integer centers)**.** *Suppose $A \in \mathbb{Z}$, $A \leq -17$, and take the odd branch of* (2.5)*. Put $N = -A - 1$. There is a unique integer $k \geq 4$ with*

$$k^2 \leq N \leq k^2 + 2k.$$

Set

$$r = k + \tfrac{1}{2}, \qquad s = c - r^2 = N - k(k+1). \tag{3.3}$$

Then $s \in \mathbb{Z}$,

$$-r + \tfrac{1}{2} \leq s \leq r - \tfrac{1}{2}, \tag{3.4}$$

and every coordinate again has the unique representation (3.2)*.*

Proof. Here $c = N + 1/4$. The integer intervals $[k^2, k^2 + 2k]$ are adjacent and disjoint because the next lower endpoint is $(k+1)^2 = k^2 + 2k + 1$. Since $N \geq 16$, we have $k \geq 4$ and $r \geq 9/2$. Formula (3.4) is just $-k \leq s \leq k$.

The maximum R and center r both belong to $\mathbb{Z} + 1/2$. If $s \leq -2$, then $c + 1 < r^2$, so Lemma 2.3 gives $R < r + 1$, hence $R \leq r$. If $s \geq -1$, the upper bound in (3.4) yields $c + 1 < (r+1)^2$. Thus $R < r + 2$, and $R \leq r + 1$.

Were $|u_i| \leq r - 2$, the first case would imply

$$u_i^2 - c \leq -4r + 4 - s \leq -3r + \tfrac{7}{2} < -2r \leq -2R.$$

The second case would imply

$$u_i^2 - c \leq -4r + 5 < -2r - 2 \leq -2R.$$

Both strict inequalities hold for $r \geq 9/2$, and both conclusions contradict (2.7). The absolute coordinates have the same half-integral parity as r. Therefore $|u_i| \in \{r-1, r, r+1\}$, giving (3.2) uniquely. $\square$

The next step is an exact arithmetic separation, not a comparison of leading terms as r tends to infinity.

Proposition 3.3 (The two local equations)**.** *In either Lemma 3.1 or Lemma 3.2, the symbols satisfy*

$$\varepsilon_{i-1} + \varepsilon_{i+1} = 2\varepsilon_i \delta_i, \tag{3.5}$$

$$\delta_{i-1} + \delta_{i+1} = \delta_i^2 - s. \tag{3.6}$$

In particular, $s \in \{-2, -1, 0, 1, 2, 3\}$.

Proof. Substitute (3.2) and $c = r^2 + s$ into the centered recurrence. After cancelling r^2, the identity is

$$r(\varepsilon_{i-1} + \varepsilon_{i+1} - 2\varepsilon_i\delta_i) = \delta_i^2 - \delta_{i-1} - \delta_{i+1} - s. \tag{3.7}$$

The coefficient multiplying r is an even integer. The expression $\delta_i^2 - \delta_{i-1} - \delta_{i+1}$ lies between -2 and 3. In the integer-center case, the right-hand side of (3.7) therefore lies in $[-r-2, r+2]$, with absolute value strictly less than $2r$. In the half-integer case it lies in $[-r-3/2, r+5/2]$, again strictly between $-2r$ and $2r$. If the even coefficient on the left were nonzero, its absolute value would be at least two, contradicting these bounds. Both sides vanish, giving (3.5)–(3.6). The range of s follows by solving (3.6) for s. □

4 The local periodic words

We now classify cyclic solutions of (3.5)–(3.6) without reference to the size or parity of the center r.

Lemma 4.1 (Complete symbol classification)**.** *Let $(\varepsilon_i, \delta_i)$ be a nonempty cyclic sequence with $\varepsilon_i \in \{-1, 1\}$ and $\delta_i \in \{-1, 0, 1\}$ satisfying (3.5)–(3.6) for an integer s. For the coordinates $u_i = \varepsilon_i r + \delta_i$, all possibilities are repetitions and cyclic rotations of the following words:*

$$\begin{array}{c|l} s & \textit{centered coordinate words} \\ \hline -1 & (1-r),\ (1+r) \\ 0 & (-r,-r,r,r) \\ 1 & (-r-1,r,r),\ (r-1,-r,-r) \\ 3 & (-r-1,r-1). \end{array} \tag{4.1}$$

There are no solutions for any other s.

Proof. Equation (3.5) has three consequences. At an offset $\delta_i = 1$, both neighboring signs equal ε_i. At an offset $\delta_i = -1$, both neighboring signs equal $-\varepsilon_i$. At an offset $\delta_i = 0$, the neighboring signs are opposite. In particular, adjacent offsets 1 and -1 are impossible: the first would make the adjacent signs equal, while the second would make them opposite. This argument also applies across the cyclic boundary and when the period is one or two.

The range assertion in Proposition 3.3, which also follows directly from (3.6), leaves only six cases.

Case $s = -2$. Equation (3.6) requires $\delta_i^2 + 2 \le 2$ at every index, so all offsets vanish. Substitution then gives $0 = 2$, which is impossible.

Case $s = -1$. An offset -1 would require both neighbors to have offset 1, contradicting the forbidden adjacency. If a zero offset occurred, its neighbors would sum to 1, so one of them would be 1. At that neighboring offset, equation (3.6) would require both adjacent offsets to be 1, contrary to the original zero. Thus all offsets are 1. Equation (3.5) makes all signs equal, giving the two constant words $(1-r)$ and $(1+r)$.

Case $s = 0$. An offset -1 would require neighboring offsets 0 and 1, forbidden by adjacency. Among the remaining values 0 and 1, a zero offset has two zero neighbors, whereas an offset 1 requires neighbors 0 and 1. An offset 1 would therefore be adjacent to a zero that cannot neighbor it. All offsets are zero, and the sign equation becomes $\varepsilon_{i+1} = -\varepsilon_{i-1}$. Its words repeat $(u, v, -u, -v)$, where $u, v \in \{-1, 1\}$. The four choices of (u, v) are cyclic rotations of $(-1, -1, 1, 1)$, giving the single centered word in (4.1).

Case $s=1$**.** If an offset 1 occurred, its neighbors would sum to zero. Neither could be -1, so both would be zero. But a zero adjacent to 1 would require its other neighbor to be -2, impossible. The offsets lie in $\{-1,0\}$. Each -1 has two zero neighbors, and each zero has one neighbor -1 and one zero. The offsets must therefore repeat $(-1,0,0)$. Starting with sign u at the offset -1, the sign equation forces the following signs to be $-u,-u,u$. The signs have the same period three. The choices $u=-1$ and $u=1$ give, respectively, $(-r-1,r,r)$ and $(r-1,-r,-r)$.

Case $s=2$**.** An offset 1 would require neighbors -1 and 0, again forbidden. A zero offset would have two neighbors with offset -1. The sign rule at each neighbor forces its sign to be the negative of the center sign, so the two neighboring signs agree. The sign rule at the center zero requires them to be opposite. Thus neither 1 nor 0 can occur. If all offsets were -1, equation (3.6) would read $-2=-1$, also impossible.

Case $s=3$**.** A zero offset would require its neighbors to sum to -3, impossible. Each remaining offset has two neighbors -1, so offset 1 is forbidden. All offsets equal -1, and the signs alternate. This gives $(-r-1,r-1)$.

These cases exhaust every cyclic solution, with no restriction on its initially specified period. □

For integer centers, $c=r^2+s=-A$, and the four allowed values of s give Table 1 for every $A\le -13$. For the odd branch, $r=k+1/2$, $x_i=u_i-1/2$, and

$$A=-k(k+1)-1-s. \tag{4.2}$$

The same words give Table 2 for every $A\le -17$. To include small parameters in the existence assertion, one must verify the words without using either large-parameter threshold.

Lemma 4.2 (Existence and exact periods)**.** *Every entry in Tables 1 and 2 is an orbit for its stated parameter and has the stated exact period. The within-row coincidences are precisely those stated in the tables.*

Proof. The centered recurrence can be checked for arbitrary real r. For $c=r^2-1$, the constant coordinates satisfy

$$(1\pm r)^2-c=2(1\pm r).$$

For $c=r^2+3$, the two-cycle identities are

$$(-r-1)^2-c=2(r-1),\qquad (r-1)^2-c=2(-r-1).$$

For $c=r^2+1$, the first three-cycle satisfies

$$(-r-1)^2-c=2r,\qquad r^2-c=-1=(-r-1)+r,$$

and the second satisfies

$$(r-1)^2-c=-2r,\qquad (-r)^2-c=-1=(r-1)+(-r).$$

For $c=r^2$, every squared coordinate in $(-r,-r,r,r)$ minus c is zero, as is the sum of its two neighbors. These identities prove existence for all listed integer values $r=k$, or half-integer values $r=k+1/2$ followed by the shift $-1/2$.

In the even branch the two-cycle entries differ by $2k$, so they are distinct for $k\ge 1$. Neither three-cycle word is constant at any integer $k\ge 0$; their two orbits coincide when $k=0$. For $k\ge 1$, the first three-cycle has a repeated positive coordinate k, whereas the second has a repeated negative coordinate $-k$, so they are not cyclic rotations. The four-cycle has neither period one nor period two for $k \geq 1$. The two fixed words coincide only at $k = 0$.

In the odd branch the two-cycle entries differ by $2k+1$, which is nonzero for $k \geq 0$. The first three-cycle is always nonconstant. The second is constant exactly at $k = 0$; its constant value is -1, which is the fixed point already listed at $A = -2$. For $k \geq 1$, the repeated coordinates in the two three-cycles are k and $-k-1$, so the orbits are distinct. The four-cycle has neither period one nor two because $-k-1 \neq k$. Finally, $-k$ and $k+1$ are always distinct fixed coordinates. A nonconstant three-letter word has least period three, completing the checks. □

5 The finite complement and completeness

The preceding argument covers every even normalized parameter $A \leq -13$ and every odd normalized parameter $A \leq -17$. The remaining cases are finite for a proved reason. We give their complete certificates, including an exact procedure for checking the point sets rather than only the number of points.

Lemma 5.1 (Finite partial permutations)**.** *Let $F : X \to X$ be injective, and suppose every periodic point of F belongs to a specified finite set $V_0 \subset X$. Define*

$$V_{j+1} = \{p \in V_j : F(p) \in V_j\}. \tag{5.1}$$

This decreasing sequence stabilizes at a set V_*, which is exactly the periodic set of F in X.

Proof. Every periodic orbit contained in V_0 is contained in each V_j by induction. A decreasing sequence of finite sets stabilizes, and the restriction $F : V_* \to V_*$ is injective. It is therefore a permutation of V_*, so every point of V_* is periodic. □

5.1 The thirteen even parameters

For $c = 0, \ldots, 12$, so that $A = -c$, use the integer quantities

$$B_c = 1 + \lfloor \sqrt{1+c} \rfloor, \qquad S_c = \{u \in \mathbb{Z} : |u| \leq B_c,\ |u^2 - c| \leq 2B_c\}. \tag{5.2}$$

Lemma 2.3 shows that every periodic coordinate belongs to S_c. Start with $V_0 = S_c \times S_c$, and apply Lemma 5.1 to

$$F_c(x, y) = (y, y^2 - c - x). \tag{5.3}$$

The bound and alphabet use only an integer square root and integer comparisons. Table A.1 gives every successive cardinality up to stabilization and every stable coordinate word.

5.2 The seventeen odd parameters

For $A = 0, -1, \ldots, -16$, use doubled centered coordinates $z_i = 2x_i + 1$, which are odd integers. The bound in (2.7) gives

$$|z_i| \leq 2 + \sqrt{1 - 4A}.$$

Define

$$\begin{aligned} &B_A = 2 + \lfloor \sqrt{1 - 4A} \rfloor, \\ &S_A = \{z \in \mathbb{Z} : z \text{ odd},\ |z| \leq B_A,\ |z^2 + 4A + 3| \leq 4B_A\}. \end{aligned} \tag{5.4}$$

Indeed, the doubled recurrence is

$$2(z_{i-1} + z_{i+1}) = z_i^2 + 4A + 3, \tag{5.5}$$

which proves the second alphabet constraint. On pairs of odd integers the map is

$$G_A(z, w) = \left(w, \frac{w^2 + 4A + 3}{2} - z \right). \tag{5.6}$$

For odd w, the integer $w^2 + 4A + 3$ is divisible by four. The quotient by two is even, so the second coordinate of G_A is odd. Solving for z gives an inverse on the odd lattice, in particular injectivity. Take $V_0 = S_A \times S_A$ and use (5.1). Table A.2 records the entire calculation; its words are expressed in the original normalized integer coordinates $x = (z - 1)/2$.

5.3 What the certificates assert

For each row of the appendix tables, let W be the displayed set of coordinate words. Define the corresponding pair set by

$$E(W) = \bigcup_{(v_0, \ldots, v_{d-1}) \in W} \{(v_i, v_{i+1}) : 0 \le i < d\}, \tag{5.7}$$

where $v_d = v_0$. Each even row asserts $V_* = E(W)$. Each odd row asserts

$$V_* = \{(2x + 1, 2y + 1) : (x, y) \in E(W)\}.$$

Thus the claimed final set is specified explicitly, not inferred from its cardinality.

For completeness, the pruning calculation in every row is the following exact finite procedure. Here S and F mean the appropriate alphabet and map just defined.

```
V = {(x, y) : x in S and y in S}
sizes = [cardinality(V)]
repeat:
    U = {p in V : F(p) in V}
    if U = V:
        return V, sizes
    V = U
    append cardinality(V) to sizes
```

All sets consist of integer pairs. The division in G_A is exact, as proved above. The last cardinality listed in a table row is unchanged by one further step; a final zero denotes the empty stable set. Expanding the words by (5.7) gives the full-set comparison for each returned V. The starting sets have at most 81 elements, so the certificate is finite without an orbit-period cutoff.

The complete point sets were also verified by a separate construction: in the original integer coordinates, form the entire bounding square or rectangle, retain exactly those one-step edges that stay within it, and compute Boolean transitive closure. A vertex is on a nonempty return path precisely when it is periodic. This second construction does not use the filtered alphabet or the pruning operation, and the comparison includes all vertices and cycle words, not just totals. The computational part is limited to the stated thirteen and seventeen parameters; the all-parameter argument is Section 3 and Lemma 4.1.

Completion of the classification in Theorem 1.1. Lemma 2.1 first reduces rational periodic points to integer points, and Lemma 2.2 reduces the coefficients to the two normal forms. Equations (2.3) and (2.4) settle the upper parameter ranges. The two six-symbol lemmas and Lemma 4.1 settle every $A \le -13$ in the even branch and every $A \le -17$ in the odd branch. The exact certificates in Tables A.1 and A.2 settle the complementary parameters. Every word there occurs in Tables 1 or 2, and Lemma 4.2 proves existence and exact periods for all entries of those tables. This proves the complete orbit lists and the period restriction. The count and equality locus are proved next. □

6 Parameter intersections and the sharp bound

The tables list parameter families, not disjoint parameter sets. We now determine all intersections, retaining the small-parameter degeneracies from Lemma 4.2.

Proposition 6.1 (Even branch)**.** *For $H_{A,0}$, the only intersections between distinct nondegenerate rows of Table 1 occur at $A = -1$ and $A = -4$. At $A = -1$, there is one three-cycle and one four-cycle; at $A = -4$, there is one two-cycle and one four-cycle. The rational periodic set has at most seven points, with equality exactly at $A = -1$.*

Proof. Use $c = -A$. The parameter families for fixed points, two-cycles, three-cycles, and four-cycles are, respectively,

$$k^2 - 1\ (k \ge 0), \quad k^2 + 3\ (k \ge 1), \quad k^2 + 1\ (k \ge 0), \quad k^2\ (k \ge 1).$$

An intersection reduces to a difference of two integer squares equal to 1, 2, 3, or 4. For $u > v \ge 0$, factor $u^2 - v^2 = (u-v)(u+v)$. The positive factor pairs show that difference 1 has only $(u, v) = (1, 0)$, difference 2 is impossible, difference 3 has only $(2, 1)$, and difference 4 has only $(2, 0)$.

A fixed-point row cannot meet a two-cycle row, since their difference 4 would force the two-cycle index to be zero. It cannot meet a three-cycle row, since that would require difference 2, or a four-cycle row, since difference 1 would force the four-cycle index to be zero. The two- and three-cycle rows are disjoint by difference 2. The two- and four-cycle rows meet only at $c = 4$, by difference 3. The three- and four-cycle rows meet only at $c = 1$, by difference 1.

At $c = 1$, the two three-cycle words coincide at their index zero, so they contribute three points, and the four-cycle contributes four. At $c = 4$, the two-cycle and four-cycle contribute two and four points. At any other three-cycle parameter there are six points, and all remaining parameters have at most four. This proves the bound and its equality case. □

Proposition 6.2 (Odd branch)**.** *For $H_{A,1}$, the only intersections between distinct period families in Table 2 occur at $A = -2, -4, -6$, where the numbers of rational periodic points are respectively 5, 8, and 4. The rational periodic set has at most eight points, with equality exactly at $A = -4$.*

Proof. Put $P(k) = k(k+1)$ for $k \ge 0$. These numbers are even and strictly increasing. The negatives of the four parameter families are $P(k)$, $P(k) + 4$, $P(k) + 2$, and $P(k) + 1$. The last is odd and cannot meet any of the first three. The two three-cycle rows are treated here as one period family, with their distinct ranges retained.

For $k > \ell \ge 0$,

$$P(k) - P(\ell) = (k - \ell)(k + \ell + 1). \tag{6.1}$$

The only solution with difference 2 is $(k, \ell) = (1, 0)$, and the only solution with difference 4 is $(2, 1)$. Indeed, the two factors are positive, the second is larger than the first, and their difference is $2\ell + 1$; the positive factor pairs of 2 and 4 leave precisely these cases.

The fixed-point and three-cycle families meet only at $A=-2$: the fixed-point index is 1 and the three-cycle index is 0. The latter contributes only its first, nonconstant word, giving $2+3=5$ points. The fixed-point and two-cycle families meet only at $A=-6$, with indices 2 and 1, giving $2+2=4$ points. The three-cycle and two-cycle families meet only at $A=-4$, with indices 1 and 0, giving $3+3+2=8$ points. These three distinct parameter values rule out a triple intersection. Outside them there are at most six points. □

Propositions 6.1 and 6.2, together with the translation (2.2), prove (1.4) and its equality locus. In particular, the normalized map attaining eight points is

$$H_{-4,1}(x,y)=(y,y^2+y-4-x).$$

Its three coordinate words are $(-2,-1)$, $(-3,1,1)$, and $(0,-2,-2)$. Their lengths are 2, 3, and 3, respectively, and expanding them into adjacent pairs gives eight distinct points. Every map on the stated equality locus is its integer translate, and there are no other equality cases.

7 Return counts and scope

The complete orbit lists also determine the rational-point fixed sets at every ordinary time. Let $N_d=N_d(a,b)$ be the number of distinct orbits of exact length d, for $d\in\{1,2,3,4\}$, obtained from the appropriate table after removing its indicated degeneracies. Put $N_d=0$ for other positive integers d.

Corollary 7.1 (Ordinary-time returns)**.** *For every $a,b\in\mathbb{Z}$ and $n\geq 1$,*

$$\#\operatorname{Fix}(H_{a,b}^n,\mathbb{Q})=N_1+2N_2\mathbf{1}_{2|n}+3N_3\mathbf{1}_{3|n}+4N_4\mathbf{1}_{4|n}. \tag{7.1}$$

The rational-point Artin–Mazur series is

$$\zeta_{a,b,\mathbb{Q}}(t):=\exp\left(\sum_{n\geq 1}\frac{\#\operatorname{Fix}(H_{a,b}^n,\mathbb{Q})}{n}t^n\right)=\prod_{d=1}^{4}(1-t^d)^{-N_d}. \tag{7.2}$$

The equality holds as a formal power-series identity, and as an analytic identity for $|t|<1$.

Proof. A point on an orbit of exact length d is fixed by the n-th iterate exactly when $d\mid n$. Each such orbit contributes d points, proving (7.1). For each d,

$$\sum_{j\geq 1}\frac{d}{dj}t^{dj}=\sum_{j\geq 1}\frac{t^{dj}}{j}=-\log(1-t^d).$$

There are only finitely many orbits. Substitution and exponentiation give (7.2) without any exchange of infinite orbit sums. □

For the equality locus in Theorem 1.1, one has $N_2=1$, $N_3=2$, and $N_1=N_4=0$, so

$$\zeta_{a,b,\mathbb{Q}}(t)=\frac{1}{(1-t^2)(1-t^3)^2}.$$

This is a rational-point orbit identity. It does not count all algebraic fixed points, fixed-scheme lengths, or local intersection multiplicities.

The proof identifies the source of the uniform bound within this specific family. Integrality gives a discrete coordinate lattice; the maximum-coordinate estimate then leaves six symbols; and the coefficient parity separates the recurrence into two rigid local equations. The integer and half-integer branches share this mechanism, but their exceptional parameters and orbit coincidences differ.

The assumptions are not removed by the elementary nature of the argument. For nonintegral rational coefficients the p-adic maximum proof no longer supplies integrality, while other leading coefficients or the opposite determinant sign change the recurrence. We make no classification claim in those settings. In particular, the fixed Jacobian-one result neither proves nor refutes the finite-exception Jacobian conjecture in [1, Section 11]. The broader rational-parameter and number-field problems remain outside the theorem.

Finally, the finite product (7.2) carries only the ordinary return data of the rational periodic set. No identification with target Euler factors, functional-equation root numbers, automorphic data, zeros of an arithmetic function, or a Hilbert–Pólya realization is asserted. The classification and the sharp eight-point bound are the conclusions established here.

A Exact finite certificates

The following tables provide the finite complement used in Section 5. Their initial alphabets and maps are defined in (5.2)–(5.3) and (5.4)–(5.6). Each cardinality list starts at V_0 and ends at V_*; the last set is unchanged by one additional application of (5.1). A zero last entry means the stable set is empty. The coordinate words specify every stable point through (5.7), including the doubling convention for the odd branch. Thus the tables describe point sets, not just cycle counts.

Table A.1: Complete even-branch certificates, $c = -A = 0, \ldots, 12$. The words are in normalized integer coordinates.

c	B_c	Successive cardinalities	Stable coordinate words
0	2	$25, 15, 10, 8, 7, 6, 5, 4, 3, 2$	(0); (2)
1	2	$25, 18, 14, 12, 10, 9, 8, 7$	$(-1, 0, 0)$; $(-1, -1, 1, 1)$
2	2	$25, 17, 12, 8, 7, 6$	$(-2, 1, 1)$; $(-1, -1, 0)$
3	3	$49, 28, 18, 11, 8, 6, 4, 2$	(-1); (3)
4	3	$49, 29, 18, 14, 12, 10, 8, 6$	$(-2, 0)$; $(-2, -2, 2, 2)$
5	3	$49, 26, 16, 9, 8, 7, 6$	$(-3, 2, 2)$; $(-2, -2, 1)$
6	3	$49, 23, 12, 6, 2, 0$	None
7	3	$36, 12, 6, 3, 2$	$(-3, 1)$
8	4	$81, 33, 14, 6, 2$	(-2); (4)
9	4	$64, 30, 14, 12, 10, 8, 6, 4$	$(-3, -3, 3, 3)$
10	4	$36, 20, 12, 6$	$(-4, 3, 3)$; $(-3, -3, 2)$
11	4	$36, 12, 6, 4, 2, 0$	None
12	4	$36, 4, 2$	$(-4, 2)$

Table A.2: Complete odd-branch certificates, $A = 0, -1, \ldots, -16$. Pruning is carried out in odd doubled coordinates, but the words shown are in normalized integer coordinates $x = (z - 1)/2$.

A	B_A	Successive cardinalities	Stable coordinate words
0	3	$16, 8, 5, 3, 2$	(0); (1)
-1	4	$16, 12, 9, 8, 7, 6, 5, 4$	$(-1, -1, 0, 0)$
-2	5	$36, 22, 17, 13, 10, 8, 7, 6, 5$	(-1); (2); $(-2, 0, 0)$
-3	5	$36, 24, 17, 14, 12, 10, 8, 6, 5, 4$	$(-2, -2, 1, 1)$
-4	6	$36, 22, 14, 8$	$(-2, -1)$; $(-3, 1, 1)$; $(-2, -2, 0)$
-5	6	$36, 20, 12, 8, 5, 2, 1, 0$	None
-6	7	$64, 32, 18, 11, 8, 5, 4$	(-2); (3); $(-3, 0)$
-7	7	$64, 32, 18, 14, 10, 8, 6, 4$	$(-3, -3, 2, 2)$
-8	7	$64, 28, 14, 7, 6$	$(-4, 2, 2)$; $(-3, -3, 1)$
-9	8	$64, 24, 10, 4, 2, 0$	None
-10	8	$36, 6, 2$	$(-4, 1)$
-11	8	$36, 8, 2, 0$	None
-12	9	$64, 22, 10, 4, 2$	(-3); (4)
-13	9	$36, 20, 14, 12, 10, 8, 6, 4$	$(-4, -4, 3, 3)$
-14	9	$36, 20, 12, 6$	$(-5, 3, 3)$; $(-4, -4, 2)$
-15	9	$36, 12, 6, 4, 2, 0$	None
-16	10	$36, 4, 2$	$(-5, 2)$

For an independent whole-box reconstruction of these same finite sets, the even branch can use

$[-B_c, B_c]^2 \cap \mathbb{Z}^2$ directly. In the odd branch, the original integer coordinates lie in

$$\left[\left\lceil \frac{-B_A - 1}{2} \right\rceil, \left\lfloor \frac{B_A - 1}{2} \right\rfloor\right]^2 \cap \mathbb{Z}^2.$$

For each of these boxes, form the directed adjacency matrix of the original map $H_{A,e}$, retaining an edge only when its endpoint remains in the box. Its Boolean transitive closure has a nonzero diagonal entry at exactly the periodic vertices. This supplies a second fully specified finite method for checking the listed complete sets without the filtered alphabets or a maximum-period guess.

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